

Scalability & Future Plan

Date	15 March 2024
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Local Deployment	The application is currently accessible only through the local system and cannot serve remote users.	High
2	Limited Training Dataset	Model performance depends on the size, quality, and diversity of the dataset used for training.	Medium
3	Single Applicant Prediction	The current application processes one applicant at a time and does not support batch prediction.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Flask application runs locally and supports limited users.	Deploy the application on a scalable cloud platform to support multiple concurrent users.
2	Data Storage	The application does not currently use a dedicated database for storing applicant and prediction records.	Integrate a secure database to manage applicant information and prediction history.
3	Performance	Predictions are generated using the locally stored Random Forest model.	Optimize the model and application backend and use scalable cloud infrastructure for improved performance.
4	Security	The current application provides basic form input handling without advanced authentication or access control.	Implement user authentication, role-based access control, secure data handling, and improved input validation.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud Deployment	1–2 Months	Make the application remotely accessible and improve scalability.
Phase 2	Database Integration	2–3 Months	Securely store applicant details, prediction records, and application history.
Phase 3	Batch Prediction & Model Enhancement	3–6 Months	Process multiple applications efficiently and improve prediction performance using larger datasets and model optimization.
Phase 4	Explainable AI & Advanced Security	6–12 Months	Provide transparent prediction explanations and improve application security, authentication, and access control.